

Soft Spaces in Low-Qubit Hilbert Spaces – Phase 2

Dimensional Continuation, Perturbative Effective-Subspace Geometry,
Controlled-Lift Theorems, 12-Qubit Confirmation, and a Falsification-Driven
Audit of Sparse-Pauli High-Dimensional Recurrence

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Abstract

Phase 2 of the Soft Spaces project was designed to test whether recurrent low-qubit structures identified in Phase 1 persist, transform, or fail when the Hilbert-space dimension is enlarged and the numerical evidence is exposed to progressively stronger physical and mathematical falsification tests. The work developed from targeted 8Q-to-9Q continuation and perturbative subspace diagnostics through 10Q, 11Q, and a predeclared checkpointed 12Q confirmation. The central high-Q score was formulated in terms of a regularized Feshbach-type effective operator on an adjacent two-eigenvector subspace,

$$\Sigma_i^{(\eta)} = P_i V Q_i g_\eta(E_i - Q_i H Q_i) Q_i V P_i, \quad g_\eta(x) = \frac{x}{x^2 + \eta^2},$$

followed by a normalized coherence ratio, a spectrum-matched REAL-versus-NULI contrast, a minimum over dephasing-like and transverse perturbation families, and a local-control prominence statistic.

The empirical Phase-2 track produced multiple positive results. Independent 9Q perturbation tests showed recurrent advantages in spectral susceptibility and projector retention for several frozen candidate coordinates. Later dimensional-transfer experiments produced a simple arithmetic continuation rule

$$k_{q+1} = 2k_q + 1,$$

which was frozen before the final 12Q run. The checkpointed 12Q test used 40 seeds in eight independent five-seed batches, three perturbation repetitions, two perturbation families, fixed local controls, and a predeclared positive-prominence gate of 0.75. Both frozen 12Q predictions passed exactly at $6/8 = 0.75$.

A second part of Phase 2 then asked a stricter question: what, precisely, had been demonstrated? An exact non-circular ancilla-lift theorem was proved for a controlled coupled model, showing preservation of the complete prominence functional under a decoupled branch lift. Perturbative stability estimates, an ancilla-parity theorem eliminating the linear response, Schur-complement bounds, second-order score-curvature estimates, and outward-rounded Arb/Acb interval arithmetic produced a computer-assisted certified radius

$$\rho_{\text{cert}} = 8.9 \times 10^{-11}$$

for the frozen positive controlled-model instances. These results are mathematically substantive, but their scope is the controlled lift, not the independently redrawn high-Q ensemble.

The final audit materially changed the physical interpretation of the high-Q recurrence. A same-seed coupling audit showed that the original dimension-dependent random-Pauli Hamiltonians are not copied ancilla lifts. A sparse-Pauli algebra audit then showed that the fixed five-term ensemble develops exponentially growing spectator multiplicity, while $k \mapsto 2k + 1$ exactly preserves normalized spectral rank. Finally, a predeclared degenerate-basis rotation test

produced sign reversals in the REAL–NULL hotspot prominence while leaving the Hamiltonians invariant to approximately 10^{-14} . Therefore the original selected ordinal two-vector high-Q hotspots are not basis-independent observables of the Hamiltonian and perturbation alone.

The scientific conclusion of Phase 2 is consequently twofold. First, the project produced a rigorous effective-subspace formalism, an exact controlled dimensional-lift theorem, and a validated computer-assisted perturbative certificate. Second, the original five-term high-Q recurrence was successfully falsified as evidence for an intrinsic transported physical hotspot: its ordinal progression is structurally explained by normalized spectral-rank preservation, sparse-Pauli degeneracy, and control geometry, and the residual score is basis dependent. This negative endpoint is not a failure of the programme; it is the result of carrying the original claim through a complete falsification chain. A new physical Phase-3 model must therefore use basis-invariant subspace observables or a sufficiently rich Hamiltonian ensemble, matched boundary controls, and a new predeclared holdout protocol.

Keywords: Hilbert space; low-qubit quantum systems; effective Hamiltonians; Feshbach projection; Nakajima–Zwanzig projection; random Pauli Hamiltonians; spectrum-matched controls; spectral degeneracy; perturbation theory; computer-assisted proof; Qiskit; falsification; basis invariance.

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1 Claims, scope, and falsification hierarchy

A central requirement of this manuscript is that different levels of evidence are not conflated. Phase 2 contains four logically distinct classes of result:

1. **Observed numerical result.** A finite computation under a specified protocol produced a reproducible value or sign pattern.
2. **Exact mathematical result.** A theorem follows from stated assumptions without numerical approximation.
3. **Computer-assisted certified result.** A finite stored matrix instance satisfies a theorem under outward-rounded interval arithmetic.
4. **Falsified interpretation.** A stronger physical interpretation is rejected by an explicit counterexample or structural audit.

These categories are intentionally kept separate throughout the paper.

Table 1: Claim hierarchy used in Phase 2.

Category	Typical evidence	Example from this work
Observed	Frozen numerical output under specified seeds and controls	The two frozen 12Q candidates each produced positive prominence in 6 of 8 batches.
Exact theorem	Closed-form derivation under stated assumptions	The decoupled ancilla lift preserves the effective operator, coherence ratio, and full prominence functional.
Computer-assisted certificate	Outward-rounded interval arithmetic on stored finite matrices	The common controlled-model certified radius $\rho_{\text{cert}} = 8.9 \times 10^{-11}$.
Falsification	Counterexample or structural mechanism incompatible with the stronger claim	Degenerate-basis rotations reverse the prominence sign while reconstructing the same Hamiltonian to approximately 10^{-14} .

The strongest admissible final statement is therefore not that a universal Soft-Space transport law has been proved. The admissible statement is that Phase 2 established a rigorous controlled effective-subspace theory while the original sparse-Pauli ordinal high-Q interpretation failed a necessary basis-invariance test.

2 Scientific aim and the role of Phase 2

The Soft Spaces programme began with an operational question rather than a theorem: can a finite-dimensional Hilbert space contain recurrent regions whose local structure responds differently to perturbation or open-system evolution than a carefully matched null model? Phase 1 established an empirical 4Q–8Q evidence base and identified candidate coordinates, local ridges, anchor regions, and recurrent REAL – NULL contrasts.

Phase 2 deliberately changed the standard of evidence. The objective was no longer merely to find more high-scoring regions. The objectives became:

1. determine whether low-Q candidate structure continues into 9Q–12Q;

2. separate static recurrence from perturbative or dynamical protection;
3. freeze coordinates and thresholds before confirmation runs;
4. use independent seed blocks and local falsification controls;
5. translate the numerical score into operator mathematics;
6. construct an exact dimensional-lift baseline that does not encode hotspot labels by hand;
7. derive stability bounds away from that baseline;
8. test whether the original high-Q ensemble actually satisfies the hypotheses of the controlled theorem;
9. search explicitly for spectral, algebraic, indexing, and basis artefacts;
10. stop the original model if a physically decisive falsification is found.

This distinction is essential. Phase 2 should not be read as one monotonically strengthening “discovery”. It is a sequence of increasingly restrictive questions. Some were confirmed; some were only conditionally proved; and the strongest original physical interpretation was ultimately rejected.

3 Hilbert-space scaling and experimental objects

For q qubits,

$$\mathcal{H}_q \cong (\mathbb{C}^2)^{\otimes q}, \quad d_q = \dim \mathcal{H}_q = 2^q.$$

The Phase-2 target dimensions therefore include

$$d_8 = 256, \quad d_9 = 512, \quad d_{10} = 1024, \quad d_{11} = 2048, \quad d_{12} = 4096.$$

The high-Q numerical model used a sparse Hermitian random-Pauli Hamiltonian of the form

$$H_q = \sum_{\ell=1}^s a_\ell P_\ell, \quad s = 5,$$

where the P_ℓ are q -qubit Pauli strings and the coefficients and Pauli labels are seed generated. The ordered spectral decomposition is

$$H_q = \sum_{n=0}^{d_q-1} \lambda_n |u_n\rangle \langle u_n|, \quad \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_{d_q-1}.$$

A candidate adjacent spectral pair beginning at ordinal i is represented by

$$P_i = |u_i\rangle \langle u_i| + |u_{i+1}\rangle \langle u_{i+1}|, \quad Q_i = \mathbb{I} - P_i,$$

with pair energy

$$E_i = \frac{\lambda_i + \lambda_{i+1}}{2}.$$

The pair is admitted by the frozen neighbouring-gap gate when

$$|\lambda_{i+1} - \lambda_i| < \varepsilon_{\text{neighbor}}, \quad \varepsilon_{\text{neighbor}} = 0.05$$

in the later high-Q confirmation protocol.

4 REAL–NULL design

The REAL-versus-NULL construction is central because it attempts to separate spectral effects from eigenbasis structure.

For each REAL Hamiltonian,

$$H_{\text{REAL}} = U_{\text{REAL}} \Lambda U_{\text{REAL}}^\dagger,$$

the NULL model retains the same ordered eigenvalue multiset Λ but replaces the eigenbasis by a seeded Haar-random basis,

$$H_{\text{NULL}} = U_{\text{NULL}} \Lambda U_{\text{NULL}}^\dagger.$$

Thus

$$\text{spec}(H_{\text{NULL}}) = \text{spec}(H_{\text{REAL}}),$$

while the basis geometry is randomized. The same perturbation family is then applied to both models. A positive REAL – NULL contrast is therefore not, by construction, attributable merely to a different one-dimensional spectrum.

This design was scientifically useful, but the later basis-invariance audit shows an important limitation: if the REAL Hamiltonian itself has degenerate eigenspaces and the score selects particular eigenvectors inside those degenerate spaces, the diagonalization convention can become part of the measured quantity. Spectrum matching alone does not cure that issue.

5 Perturbation families and effective-subspace operator

The later frozen protocol used two physically distinct perturbation families:

- dephasing-like perturbations built from local Z and nearest-neighbour ZZ terms;
- transverse perturbations built from local X and nearest-neighbour XX terms.

For a perturbation V , define the regularized scalar resolvent

$$g_\eta(x) = \frac{x}{x^2 + \eta^2}.$$

The corresponding operator-valued function is

$$g_\eta(A) = A(A^2 + \eta^2 \mathbb{I})^{-1}.$$

The second-order effective operator acting on the selected pair is

$$\Sigma_i^{(\eta)}(H, V) = P_i V Q_i g_\eta(E_i - Q_i H Q_i) Q_i V P_i$$

which is a regularized Feshbach-type self-energy. Writing

$$w_n = \begin{pmatrix} \langle u_n | V | u_i \rangle & \langle u_n | V | u_{i+1} \rangle \end{pmatrix},$$

one obtains the explicit 2×2 representation

$$\Sigma_i^{(\eta)} = \sum_{n \notin \{i, i+1\}} g_\eta(E_i - \lambda_n) w_n^\dagger w_n.$$

This formula is one of the main mathematical objects produced by Phase 2. It makes the relation between local spectral geometry, perturbative coupling, and the external Hilbert-space complement explicit.

5.1 Why the regularizer has the form $g_\eta(x) = x/(x^2 + \eta^2)$

The regularization is not inserted merely as a numerical convenience. It is the Hermitian part of a standard complex-shifted resolvent:

$$g_\eta(x) = \frac{1}{2} \left[\frac{1}{x - i\eta} + \frac{1}{x + i\eta} \right].$$

Consequently, for a self-adjoint operator A ,

$$g_\eta(A) = \frac{1}{2} \left[(A - i\eta\mathbb{I})^{-1} + (A + i\eta\mathbb{I})^{-1} \right].$$

This choice has three useful properties.

First, it removes the pole of the unregularized Feshbach resolvent when E_i approaches $\text{spec}(Q_i H Q_i)$. Second, it preserves Hermiticity, because $g_\eta(x)$ is real for real x . Third, it retains the sign of the detuning:

$$\text{sgn } g_\eta(x) = \text{sgn } x,$$

so levels below and above the selected pair contribute with opposite signs. This signed structure is precisely what makes the normalized coherence ratio sensitive to constructive retention versus cancellation.

The maximal scalar magnitude is

$$\max_x |g_\eta(x)| = \frac{1}{2\eta},$$

attained at $|x| = \eta$. Therefore

$$\|g_\eta(A)\|$$

$\leq \frac{1}{2\eta}$, which is also the starting point for the later perturbation bounds.

A different regularization, for example the full complex resolvent $(x + i\eta)^{-1}$, would retain both dispersive and absorptive parts. The present Phase-2 score deliberately uses only the Hermitian/dispersive part because the implemented matrix functional is real-valued and designed as a structural comparison rather than a decay-rate observable.

6 Normalized coherence ratio

The implemented score normalizes the norm of the signed sum by the sum of the individual contribution norms:

$$C_i(H, V) = \frac{\left\| \sum_{n \notin \{i, i+1\}} g_\eta(E_i - \lambda_n) w_n^\dagger w_n \right\|_F}{\sum_{n \notin \{i, i+1\}} |g_\eta(E_i - \lambda_n)| \|w_n^\dagger w_n\|_F}$$

whenever the denominator is non-zero.

Because

$$\|w_n^\dagger w_n\|_F = \|w_n\|_2^2,$$

the triangle inequality immediately yields

$$0 \leq C_i(H, V) \leq 1.$$

The original implementation used terminology suggestive of “cancellation”. The mathematically correct interpretation is more precise: large C_i means the signed effective contributions retain a

larger coherent resultant relative to the sum of their magnitudes; small C_i means stronger mutual cancellation.

For one family f ,

$$\Delta C_{f,i} = C_{f,i}^{\text{REAL}} - C_{f,i}^{\text{NULL}}.$$

A positive value therefore means that the REAL model retains a larger normalized effective resultant than the spectrum-matched NULL model.

6.1 Exact invariance inside the selected pair

For a unitary $U \in U(2)$ acting only inside the selected pair,

$$w_n \mapsto w_n U.$$

Then

$$\Sigma_i \mapsto U^\dagger \Sigma_i U,$$

so

$$\|\Sigma_i\|_F$$

is unchanged, and the denominator is also unchanged because $\|w_n U\|_2 = \|w_n\|_2$. Hence C_i is invariant under basis rotations *within the already selected two-dimensional subspace*.

This result must not be confused with invariance under rotations in a larger degenerate eigenspace. The distinction becomes decisive in the final audit.

7 From branch scores to hotspot prominence

The later Phase-2 protocol used multiple robustification layers.

Let $f \in \{Z, X\}$ label the two perturbation families, r the perturbation repetition, b the independent batch, and k the frozen candidate coordinate.

First,

$$\Delta C_{f,k,b,r} = C_{f,k,b,r}^{\text{REAL}} - C_{f,k,b,r}^{\text{NULL}}.$$

The repetition median is

$$D_{f,k,b} = \text{median}_r \Delta C_{f,k,b,r}.$$

The robust two-family score is

$$R_{k,b} = \min\{D_{Z,k,b}, D_{X,k,b}\}.$$

For the frozen local-control neighbourhood $N_q(k)$, formed from offsets within ± 10 in steps of 2, the background is

$$B_{k,b} = \text{median}_{c \in N_q(k)} R_{c,b}.$$

The batch prominence is then

$$\boxed{\Pi_{k,b} = R_{k,b} - B_{k,b}}$$

and the empirical positive-prominence rate is

$$\hat{p}_k = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\Pi_{k,b} > 0\}.$$

For the final 12Q confirmation,

$$B = 8, \quad \hat{p}_k \geq 0.75$$

was frozen as the decision gate.

8 Early Phase-2 continuation: 8Q to 9Q

The early Phase-2 track developed through a sequence of increasingly physical diagnostics rather than one final score.

8.1 Targeted continuation and local ranking

The baseline first extended the Phase-1 candidate geometry into 9Q and later 10Q–12Q target zones. Candidate selection was deliberately separated from later confirmation. Local ranking combined structural recurrence, candidate/anchor proximity, static separation, perturbative behaviour, and branch consistency, with the explicit intention that the scoring formula should not be refitted after seeing the confirmation output.

8.2 Projector-overlap tracking

A key methodological correction was to track the perturbed two-dimensional subspace by maximum projector overlap instead of merely choosing the closest energy pair after perturbation.

If P_0 is the original rank-two projector and P_η a candidate perturbed projector, the overlap statistic can be written as

$$\mathcal{O}(P_0, P_\eta) = \frac{1}{2} \text{Tr}(P_0 P_\eta), \quad 0 \leq \mathcal{O} \leq 1.$$

The favourable contrast was

$$\Delta\mathcal{O} = \mathcal{O}_{\text{REAL}} - \mathcal{O}_{\text{NULL}} > 0.$$

In parallel, a spectral-susceptibility quantity χ was used with the favourable convention

$$\Delta\chi = \chi_{\text{NULL}} - \chi_{\text{REAL}} > 0,$$

meaning that the REAL pair is less susceptible to the perturbation than its matched NULL counterpart.

8.3 Independent 9Q confirmation

One frozen 9Q confirmation used two independent seed blocks of 25 seeds each, five perturbation strengths,

$$\eta \in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\},$$

three perturbation repetitions per seed, and five perturbation terms.

The tested source coordinates included

$$192, 0, 157, 160, 165, 67, 251.$$

The predeclared descriptive threshold classified a candidate as STRONG when both projector-overlap and susceptibility sign recurrence reached at least 0.80.

Four candidates were STRONG:

$$192, 165, 160, 157.$$

For example, candidate 192 gave overlap-positive and susceptibility-positive rates of 1.0 and 1.0, respectively, with median

$$\Delta\mathcal{O} \approx +0.0028, \quad \Delta\chi \approx +0.3273.$$

Candidate 157 gave

$$\text{Ov+} = 0.8, \quad \chi+ = 1.0,$$

with median

$$\Delta\mathcal{O} \approx +0.0020, \quad \Delta\chi \approx +0.2764.$$

These results established a real empirical perturbative signal on independent seeds. They did not yet prove that the signal identified a unique intrinsic Hilbert-space object.

9 Falsification controls, anisotropy, and local geometry

The next sequence of versions tested whether the positive perturbative effect was candidate-specific or merely common in nearby spectral regions.

The workflow included:

- matched local controls;
- local REAL-profile measurements;
- effective-subspace perturbation analysis;
- target confirmation on new seed blocks;
- multi-target generalization;
- anisotropic-direction analysis;
- high-confidence replication;
- local matrix-geometry diagnostics;
- confirmation of selected target geometry;
- path organization across the candidate neighbourhood.

The methodological principle was explicit: a candidate was scientifically stronger only if its favourable sign recurrence exceeded the behaviour of predeclared matched controls. A control performing equally well was to weaken the Soft-Space interpretation, not to be hidden.

The anisotropy work was especially important because it rejected the idea that only a scalar perturbation “strength” matters. The effective operator contains

$$PVQ g_\eta(E - QHQ) QVP,$$

so the orientation of V relative to the selected subspace and its complement is structurally relevant. Distinct perturbation families can therefore probe different matrix geometry even at comparable operator norm.

10 REAL–NULL cancellation, model transfer, and hotspot prediction

Later Phase-2 versions progressively simplified and stress-tested the mechanism.

The sequence included:

1. confirmation of a REAL – NULL difference in the matrix-level effective contribution;
2. dimensional transfer of the same cancellation/coherence law;
3. replacement of the original perturbation generator by a local/2-local chain model;
4. transfer across a second perturbation family;
5. a physics-based hotspot predictor;
6. frozen hotspot confirmation;
7. dimensional hotspot transfer;
8. blind 10Q hotspot discovery;
9. 10Q confirmation;
10. an 11Q dimensional-rule test;
11. upward-validation tests over lower dimensions;
12. harmonized 7Q–9Q validation.

The important methodological development was the move from broad descriptive features to a small, frozen perturbative mechanism. By the time of the harmonized validation, the primary statistic was no longer a large composite feature score. It was the branch-based matrix coherence contrast and local prominence defined above.

11 The dimensional rule and frozen geometry

The arithmetic continuation rule that emerged and was later frozen is

$$\boxed{k_{q+1} = 2k_q + 1.}$$

For the harmonized lower-dimensional tests, representative branch coordinates were

source $\rightarrow$ target	branch A	branch B
$6Q \rightarrow 7Q$	15	47
$7Q \rightarrow 8Q$	31	95
$8Q \rightarrow 9Q$	63	191

with the two target-space lifts of each source coordinate implemented by

$$i = k, \quad i = k + 2^{q_{\text{source}}}.$$

It is essential to distinguish three maps:

- (a) the arithmetic source-coordinate map $k \mapsto 2k + 1$;
- (b) the branch injections $i \mapsto i$ and $i \mapsto i + 2^q$;
- (c) a physical isometry $J : \mathcal{H}_q \rightarrow \mathcal{H}_{q+1}$.

Only the first two are automatically defined by indices. A physical intertwiner requires operator relations that do not follow from integer arithmetic.

12 Predeclared 12Q confirmation protocol

The final 12Q run was treated as a frozen confirmation experiment rather than an exploratory search. The scientific value of the result depends on separating decisions made before execution from observations made afterward.

The following items were fixed before the confirmation run:

Table 2: Frozen 12Q protocol.

Protocol element	Frozen value
Source coordinates	$k = 511$ and $k = 1535$.
Target-space lifts	$(511, 512)$, $(2559, 2560)$, $(1535, 1536)$, and $(3583, 3584)$.
Dimensional rule	$k_{q+1} = 2k_q + 1$.
Local controls	Source coordinate offsets within ± 10 , step 2.
Hamiltonian ensemble	Five-term random-Pauli Hamiltonian used in the high-Q protocol.
Perturbation families	Dephasing-like Z/ZZ and transverse X/XX .
Perturbation repetitions	Three repetitions per branch/seed/family instance.
Batching	Eight independent batches of five seeds each, 40 seeds total.
Neighbour-gap eligibility	$ \lambda_{i+1} - \lambda_i < 0.05$.
Branch aggregation	Median across repetitions within each perturbation family.
Two-family aggregation	Minimum of dephasing and transverse REAL – NULL contrasts.
Local background	Median robust score over the frozen local controls.
Primary statistic	Prominence $\Pi_{k,b} = R_{k,b} - B_{k,b}$.
Decision rule	Positive-prominence rate $\hat{p}_k \geq 0.75$, i.e. at least 6 of 8 batches.

The key point is methodological: the candidate coordinates, control geometry, aggregation rule, and threshold were not selected after inspection of the 12Q batch signs. The later sparse-Pauli and basis-invariance audits were separate falsification experiments and did not retroactively alter the frozen 12Q decision rule.

13 Checkpointed 12Q confirmation

The final predeclared high-Q confirmation used the source coordinates

$$k \in \{511, 1535\}.$$

The lifted target pairs were

$$\begin{aligned} k = 511 : & \quad (511, 512), (2559, 2560), \\ k = 1535 : & \quad (1535, 1536), (3583, 3584). \end{aligned}$$

The run used:

- eight independent batches;
- five seeds per batch, hence 40 seeds;
- three perturbation repetitions;

- dephasing and transverse families;
- fixed local radius ± 10 ;
- control step 2;
- 240 checkpoint records;
- predeclared decision gate $\hat{p}_k \geq 0.75$.

The result was:

Table 3: Frozen 12Q confirmation summary.

k	positive batches	rate	med. robust	med. background	med. prominence	status
511	6/8	0.750	0.073706	0.021030	0.038702	confirmed
1535	6/8	0.750	0.075264	0.029998	0.043595	confirmed

The dephasing-positive rates were 1.000 for both candidates. The transverse-positive rates were 0.875 and 0.750, respectively.

At this point the empirical statement “2/2 frozen 12Q predictions pass the predeclared gate” was correct. The remainder of Phase 2 was devoted to deciding how much physical meaning that statement could bear.

14 Qiskit translation and a gate-level witness

A Qiskit layer was added to demonstrate a route from the exact matrix calculation toward circuit-level measurement.

For eigenstates $|p\rangle, |q\rangle$, a perturbative transition amplitude is

$$B_{qp} = \langle q | \Delta H | p \rangle.$$

The design cross-checked selected complex amplitudes obtained by Qiskit statevector methods against the classical matrix calculation and constructed Hadamard-test circuits for the real and imaginary parts of representative Pauli matrix elements.

A reduced four-qubit hardware-oriented pilot was then evaluated with 4096 shots per circuit. It was explicitly not presented as the full 12Q algorithm. In ideal simulation, the normalized REAL – NULL deltas were approximately

$$+0.037231 \quad (\text{dephasing}), \quad +0.048340 \quad (\text{transverse}),$$

and both passed the frozen positive-delta gate. Under the selected noisy simulation they remained positive,

$$+0.031144, \quad +0.042338.$$

This result supports the measurability of a reduced witness. It does not convert the complete high-Q classical statistic into a presently economical hardware experiment.

15 Projection-operator bridge

The effective-subspace formalism has a natural connection to both Feshbach projection and open-system projection techniques.

Given P and $Q = \mathbb{I} - P$, the unregularized Feshbach self-energy is

$$\Sigma_P(E) = PVQ(E - QHQ)^{-1}QVP.$$

The Phase-2 quantity replaces the singular resolvent by the regularized function g_η .

On operator space, define

$$\mathcal{P}(A) = PAP, \quad \mathcal{Q} = \mathcal{I} - \mathcal{P}.$$

Then

$$\mathcal{P}^2 = \mathcal{P}, \quad \mathcal{Q}^2 = \mathcal{Q}, \quad \mathcal{P}\mathcal{Q} = \mathcal{Q}\mathcal{P} = 0.$$

For a Liouvillian $\mathcal{L}(t)$, the exact Nakajima–Zwanzig memory kernel has the structural form

$$\mathcal{K}(t, s) = \alpha^2 \mathcal{P}\mathcal{L}(t)\mathcal{G}(t, s)\mathcal{Q}\mathcal{L}(s)\mathcal{P},$$

which explicitly contains the sequence

$$\mathcal{P} \longrightarrow \mathcal{Q} \longrightarrow \mathcal{P}.$$

For a time-independent generator, the Laplace transform introduces the resolvent

$$\int_0^\infty e^{-zt} e^{t\mathcal{Q}\mathcal{L}\mathcal{Q}} dt = (z - \mathcal{Q}\mathcal{L}\mathcal{Q})^{-1},$$

when convergent. Thus the Hilbert-space effective operator and the operator-space memory kernel share a genuine projection–complement– projection architecture, while remaining mathematically different objects.

16 A deterministic prominence-stability theorem

The minimum and median operators are 1-Lipschitz in the sup norm. This gives a useful theorem that separates the mathematical target from the empirical sample.

Suppose a source target k and lifted target k' have paired control sets, and for a fixed batch

$$|D'_{f,k'} - D_{f,k}| \leq \varepsilon_T$$

for both perturbation families, while

$$|D'_{f,c'} - D_{f,c}| \leq \varepsilon_C$$

for all paired controls. Then

$$\boxed{|\Pi'_{k'} - \Pi_k| \leq \varepsilon_T + \varepsilon_C.}$$

Hence

$$\Pi_k > \varepsilon_T + \varepsilon_C \implies \Pi'_{k'} > 0.$$

If a common bound ε applies to target and controls,

$$\Pi_k > 2\varepsilon$$

is sufficient.

For eight batches, if at least six source batches satisfy the corresponding margin condition, the 0.75 positive-prominence gate is preserved.

This theorem identifies the real proof obligation: it is not enough to control the target pair. The NULL construction, both perturbation families, and the local-control background must be controlled as well.

17 Exact controlled ancilla lift

A non-circular dimensional-lift baseline can be constructed exactly.

Let

$$\mathcal{H}_{q+1} = \mathbb{C}^2 \otimes \mathcal{H}_q, \quad J_b |\psi\rangle = |b\rangle \otimes |\psi\rangle, \quad b \in \{0, 1\}.$$

Choose $\Delta > 0$ and define

$$H^\uparrow = |0\rangle\langle 0| \otimes (H - \Delta \mathbb{I}) + |1\rangle\langle 1| \otimes (H + \Delta \mathbb{I}),$$

with

$$V_f^\uparrow = \mathbb{I}_2 \otimes V_f.$$

If

$$2\Delta > \lambda_{\max}(H) - \lambda_{\min}(H),$$

the two branches do not interleave.

For

$$P_{i,b}^\uparrow = J_b P_i J_b^\dagger,$$

one obtains exactly

$$\Sigma_{i,b}^{(\eta),\uparrow} = J_b \Sigma_i^{(\eta)} J_b^\dagger$$

and therefore

$$C_{i,b}(H^\uparrow, V_f^\uparrow) = C_i(H, V_f).$$

If the REAL and NULL systems, perturbation repetitions, target, and all controls are lifted consistently, every layer of the prominence functional is preserved:

$$\Pi_{k,b}^\uparrow = \Pi_{k,b}.$$

This is an exact theorem. It does not insert 511, 1535, or any other hotspot label into the model; all adjacent pairs are preserved by the same construction. It is therefore non-circular.

Its limitation is equally exact: the original high-Q experiment did not construct H_{q+1} by copying H_q into a decoupled ancilla branch. The theorem proves a mechanism that *can* preserve prominence, not that the independently generated ensemble *does* use that mechanism.

18 Perturbation bounds away from the exact lift

The identity

$$g_\eta(A) = \frac{1}{2} \left[(A - i\eta \mathbb{I})^{-1} + (A + i\eta \mathbb{I})^{-1} \right]$$

gives the resolvent bound

$$\|g_\eta(A') - g_\eta(A)\| \leq \frac{\|A' - A\|}{\eta^2}$$

for self-adjoint A, A' , together with

$$\|g_\eta(A)\| \leq \frac{1}{2\eta}.$$

For aligned P, Q , let

$$\Sigma = PVQg_\eta(A)QVP, \quad \Sigma' = PV'Qg_\eta(A')QV'P.$$

If

$$M = \max\{\|V\|, \|V'\|\}, \quad \delta_V = \|V' - V\|, \quad \delta_A = \|A' - A\|,$$

then

$$\|\Sigma' - \Sigma\| \leq \frac{M\delta_V}{\eta} + \frac{M^2\delta_A}{\eta^2}.$$

For the normalized ratio $C = N/D$, if

$$D, D' \geq d_0 > 0, \quad |N' - N| \leq \delta_N, \quad |D' - D| \leq \delta_D,$$

then

$$|C' - C| \leq \frac{\delta_N + \delta_D}{d_0}.$$

Projector motion requires a separate cluster-gap control. If g_* is the minimum separation of the selected rank-two cluster from the remainder of the spectrum, then a sufficient isolation condition is

$$\|\delta H\| < \frac{g_*}{2}.$$

With

$$\rho = \frac{\|\delta H\|}{g_*},$$

this is simply

$$\rho < \frac{1}{2}.$$

19 Controlled-coupling stress test

A fixed structured perturbation of the exact ancilla lift was introduced as

$$H(\varepsilon) = H^\dagger + \varepsilon(X_a \otimes G),$$

with a frozen target, frozen controls, five fixed seeds, two fixed perturbation families, and a predeclared grid in the dimensionless coupling

$$\rho = \frac{\|\delta H\|}{g_*}.$$

Three of the five source seeds had positive baseline prominence; the two negative seeds were retained. Along the structurally safe grid

$$\rho \leq 0.3 < 0.5,$$

all three positive source instances satisfied the target-plus-control prominence inequality in both branches.

Directly computed prominence remained positive even through the much larger tested value $\rho = 10$, although beyond $\rho = 0.5$ pair identity is no longer guaranteed by the simple isolation theorem. This separation between a conservative sufficient theorem and a much more stable observed path became the motivation for the parity analysis.

20 Ancilla parity and the absence of linear response

Define

$$\Gamma = Z_a \otimes \mathbb{I}.$$

For

$$H(\varepsilon) = H^\dagger + \varepsilon(X_a \otimes G), \quad V^\dagger = \mathbb{I}_a \otimes V,$$

one has

$$\Gamma H(\varepsilon) \Gamma = H(-\varepsilon), \quad \Gamma V^\dagger \Gamma = V^\dagger.$$

Because the complete pair score is invariant under simultaneous unitary conjugation of H , V , and the consistently tracked projector,

$$C(\varepsilon) = C(-\varepsilon).$$

The same symmetry passes through REAL – NULL subtraction, the family minimum, the control median, and prominence:

$$\boxed{\Pi(\varepsilon) = \Pi(-\varepsilon).}$$

Hence, while the relevant branches remain isolated,

$$\boxed{\Pi(\varepsilon) = \Pi(0) + O(\varepsilon^2).}$$

This exact selection rule explains why a first-order worst-case perturbation bound can be extremely pessimistic for the controlled path.

A numerical decomposition confirmed this structure: over the frozen low-coupling grid, fitted response exponents ranged from approximately 1.999691 to 2.002899, with median 2.000099. The directly computed $+\rho$ and $-\rho$ prominences agreed at printed numerical precision.

21 Schur-complement mechanism

Write the controlled Hamiltonian in block form,

$$H(\varepsilon) = \begin{pmatrix} A & \varepsilon G \\ \varepsilon G & D \end{pmatrix}.$$

For an eigenvector $(\psi_A, \psi_D)^T$ with eigenvalue E ,

$$\psi_D = -\varepsilon(D - E)^{-1}G\psi_A,$$

whenever the inverse exists. Substitution gives

$$\left[A - E - \varepsilon^2 G(D - E)^{-1}G \right] \psi_A = 0.$$

The induced branch self-energy is therefore

$$\boxed{\Sigma_A(E, \varepsilon) = \varepsilon^2 G(D - E)^{-1}G.}$$

If Γ_{br} is the uncoupled separation of the two branch spectral intervals, then

$$\text{dist}(E, \text{spec}(D)) \geq \Gamma_{\text{br}} - |\varepsilon|$$

and

$$\boxed{\|\Sigma_A(E, \varepsilon)\|_2 \leq \frac{\varepsilon^2 \|G\|_2^2}{\Gamma_{\text{br}} - |\varepsilon|}.}$$

This is a continuum second-order statement and gives an operator-level reason for the observed quadratic response.

22 Uniform analytic and curvature certificates

A first all-directions bound produced a very small common candidate coupling radius of order

$$\rho \sim 4.24 \times 10^{-16}$$

for the three positive source instances. This was recognized as deliberately conservative, dominated by large inverse powers of the regularizer $\eta = 10^{-3}$, and not as an observed failure scale.

The parity-aware second-order programme then bounded first and second derivatives of:

- Riesz spectral projectors;
- regularized resolvents;
- the effective-operator numerator;
- the normalization denominator;
- the full ratio C ;
- target and control REAL – NULL contrasts;
- the final prominence.

For a source prominence $m = \Pi(0) > 0$ and a uniform curvature bound

$$|\Pi''(\varepsilon)| \leq K_\Pi,$$

evenness of Π gives

$$\Pi(\varepsilon) \geq m - \frac{1}{2}K_\Pi\varepsilon^2.$$

Thus a sufficient positive-sign condition is

$$|\varepsilon| < \sqrt{\frac{2m}{K_\Pi}}.$$

The double-precision continuum evaluation produced a common candidate radius of approximately

$$9.0 \times 10^{-11},$$

subject at that stage to the need for validated outward rounding.

23 Computer-assisted Arb/Acb validation

The interval-validation stage re-evaluated the frozen constant chain with Arb/Acb ball arithmetic at 256-bit working precision. The validation explicitly enclosed:

1. eigenvalues and eigenvectors of the frozen matrices;
2. rank-one and rank-two spectral gaps;
3. coupling and perturbation operator norms;
4. numerator and denominator lower bounds;
5. first- and second-derivative constants;
6. source-prominence intervals;
7. conversion from the certified ε -radius to ρ .

23.1 Implementation and reproducibility of the interval certificate

The validated stage used Arb/Acb-style ball arithmetic with outward rounding at 256-bit working precision. The certification path was deliberately separated from the Qiskit circuit layer: the interval proof concerns the stored finite matrices entering the controlled effective-subspace theorem, not shot-noise data from quantum hardware.

The validation pipeline recomputed or enclosed the quantities needed for the deterministic bound rather than trusting double-precision point estimates. The chain included:

spectral gaps $\rightarrow$ projector/resolvent bounds $\rightarrow$ numerator/denominator bounds $\rightarrow$ score curvature $\rightarrow$ prominence

The common certified radius is defined as the minimum lower bound over the positive frozen controlled instances:

$$\rho_{\text{cert}} = \min_j \underline{\rho}_j.$$

This is why the reported common radius is 8.9×10^{-11} , even though one individual positive case admits a larger lower bound.

For the three positive frozen source seeds, the validated results were:

Table 4: Outward-rounded controlled-model certificates.

seed	validated interval ρ	prominence lower	curvature upper	certified ρ lower
2533001	1.0×10^{-10}	0.140983066	$2.19972891 \times 10^{21}$	8.9×10^{-11}
2533003	3.0×10^{-10}	0.0736388287	$5.81115634 \times 10^{20}$	$1.07914365 \times 10^{-10}$
2533004	1.0×10^{-10}	0.235286838	$6.64435839 \times 10^{21}$	8.9×10^{-11}

Hence

$$\rho_{\text{cert}} = 8.9 \times 10^{-11}.$$

A precision-stability audit at 192, 256, and 384 bits reproduced the same displayed common certified lower bound. The two negative source seeds remained strictly negative and were retained.

The scope statement is crucial: this is a computer-assisted theorem for the stored finite matrices of the controlled ancilla construction. It is not a proof for an unspecified pre-rounding random distribution, nor for the independently redrawn 8Q–12Q ensemble.

24 Audit of the link to the independent high-Q ensemble

Having proved a rigorous controlled model, Phase 2 returned to the original high-Q experiment and asked whether the independent random-Pauli ensemble was actually coupled to that theorem.

A favourable same-seed 11Q–12Q audit found:

- the coefficient sequences were identical for all 40 same-seed comparisons;
- exact copied-ancilla Hamiltonians: 0/40;
- exactly copied Pauli terms: 0/200;
- Hamiltonians preserving both elementary leading-ancilla computational sectors: 0/40;
- every 12Q Hamiltonian contained at least one X or Y on the leading qubit.

Thus the direct block-ancilla intertwiner required to identify the original ensemble with the controlled theorem is absent in this frozen audit.

The correct logical separation is therefore:

$$\boxed{\text{controlled ancilla theorem} \neq \text{original independent high-Q recurrence.}}$$

25 Sparse-Pauli algebra and the spectral-artifact audit

The next audit identified a deeper structural mechanism.

25.1 Normalized spectral-rank identity

The arithmetic rule

$$k' = 2k + 1$$

obeys

$$\boxed{\frac{k' + 1}{2^{q+1}} = \frac{k + 1}{2^q}}.$$

Thus it exactly preserves the normalized cumulative spectral rank represented by the ordinal $k + 1$.

The two recurrent chains lie at normalized source positions $1/4$ and $3/4$. After the two-branch lift, the four tested target pairs occur at

$$\frac{1}{8}, \quad \frac{3}{8}, \quad \frac{5}{8}, \quad \frac{7}{8}.$$

This identity is arithmetic, not dynamical. Nevertheless, once spectral multiplicity is present it can create a highly regular apparent “continuation” of spectral boundaries.

25.2 Five-term Pauli algebra

For a fixed number $s = 5$ of Pauli generators, increasing q increases the number of spectator degrees of freedom. The 12Q audit found:

- symplectic rank 2 with forced multiplicity 256: 5/40 seeds;
- symplectic rank 4 with forced multiplicity 512: 35/40 seeds;
- a common Pauli anticommuter, implying exact $E \leftrightarrow -E$ spectral symmetry: 40/40;
- regular repeat factor

$$2^{12-5} = 128.$$

The dimension dependence of the frozen local controls was especially revealing:

Table 5: Controls forced inside repeated spectral blocks by dimension.

target	regular repeat factor	controls inside repeated block / 10
7Q	4	6/10
8Q	8	8/10
9Q	16	10/10
10Q	32	10/10
11Q	64	10/10
12Q	128	10/10

At high q , the candidates lie at algebraic spectral boundaries while the fixed ± 10 , step-2 controls increasingly lie inside exactly flat repeated blocks. Candidate and control are therefore not asymptotically of the same spectral-boundary class.

25.3 Dependence of the two frozen 12Q candidates

The local-label audit found both frozen 12Q boundaries ranked first among their 11-point local candidate/control sets. However, the two batch-prominence vectors had correlation

$$0.989390733,$$

and the candidates succeeded in the same six batches and failed in the same two batches. The eligible seed sets were identical.

Therefore the two 6/8 successes must not be treated as independent replications. For one candidate, under a simple $p = 1/2$ sign null,

$$\Pr[X \geq 6], \quad X \sim \text{Binomial}(8, 1/2),$$

equals

$$0.14453125.$$

That calculation is not a complete statistical model of the experiment, but it shows why “2/2 confirmed” cannot legitimately be read as two independent small- p events.

26 Degenerate-basis invariance as a necessary physical condition

Let E be a degenerate eigenvalue with eigenspace $\mathcal{E}_E$ of dimension $m > 1$. The Hamiltonian determines the full spectral projector

$$P_E = \sum_{j=1}^m |e_j\rangle \langle e_j|,$$

but it does not determine a preferred basis $\{|e_j\rangle\}_{j=1}^m$.

For every $U_E \in U(m)$,

$$|e'_j\rangle = \sum_{\ell=1}^m |e_\ell\rangle (U_E)_{\ell j}$$

gives the same Hamiltonian and the same P_E .

26.1 Necessary invariance condition

Let $\mathcal{F}(H, V)$ denote any proposed intrinsic hotspot observable. If H has a degenerate eigenspace $\mathcal{E}_E$, and U_E acts unitarily only inside that eigenspace, then

$$U_E H U_E^\dagger = H.$$

Therefore a necessary condition for intrinsic basis independence is

$$\boxed{\mathcal{F}(H, V) = \mathcal{F}(U_E H U_E^\dagger, U_E V U_E^\dagger)}$$

for physically equivalent representations, or, when V is held fixed as an independently defined physical operator and only the eigendecomposition basis is changed,

$$\boxed{\mathcal{F}(H, V) \text{ must not depend on the arbitrary eigenvector coordinates used inside } \mathcal{E}_E.}$$

Failure of this condition does not mean that the numerical implementation is incorrect. It means that the proposed scalar is not an intrinsic observable of the Hamiltonian-plus-perturbation pair unless an additional physical basis-selecting observable is specified.

If a score selects only the ordinal pair

$$\text{span}\{|u_i\rangle, |u_{i+1}\rangle\}$$

inside a larger degenerate eigenspace, then the selected rank-two projector is not determined by H alone. A physically intrinsic observable must therefore be invariant under all such block rotations, or it must specify an independent physical observable that selects a basis.

This is the decisive distinction between:

- invariance under $U(2)$ rotations *inside an already selected two-dimensional subspace*, which C_i has; and
- invariance under rotations in the *larger degenerate eigenspace* that alter which two-dimensional slice is selected, which the original ordinal protocol does not have in general.

27 Frozen basis-invariance falsification

The final test instantiated that issue at 8Q, where complete calculations are tractable.

Before execution it fixed:

- 12 Hamiltonian seeds;
- source coordinates 31 and 95;
- both target branches;
- local radius ± 10 , step 2;
- dephasing Z/ZZ and transverse X/XX perturbations;
- the same coherence ratio and prominence definitions;
- eight deterministic Haar rotations inside every degenerate eigenspace.

The REAL and NULL degenerate eigenbases were rotated separately. Hamiltonians, eigenvalues, perturbations, candidate coordinates, local controls, gap gate, and regularizer were held fixed.

The largest Hamiltonian reconstruction discrepancies were approximately

$$\|H'_{\text{REAL}} - H_{\text{REAL}}\|_2 = 1.434724 \times 10^{-14},$$

and

$$\|H'_{\text{NULL}} - H_{\text{NULL}}\|_2 = 9.209561 \times 10^{-15}.$$

Among ten eligible seed-candidate cases:

Table 6: Degenerate-basis falsification result.

rotation	sampled range crossed zero	baseline sign reversed
REAL degenerate basis	2/10	2/10
NULL degenerate basis	4/10	4/10

A single sign reversal is already a counterexample to basis-invariant hotspot sign. Multiple frozen counterexamples therefore establish that the original ordinal two-vector prominence is not a well-defined observable of the degenerate Hamiltonian and perturbation alone.

This should be stated in its strongest precise form:

$$\boxed{\text{basis dependence} \implies \text{the original ordinal-pair score is not intrinsic to } (H, V).}$$

The appropriate conclusion is not that the Hamiltonian hotspot “disappears” under rotation. Rather, the selected ordinal two-vector slice is not uniquely defined by the degenerate Hamiltonian, and the prominence sign can change when that arbitrary slice changes while the Hamiltonian itself remains physically the same.

28 Final logical status of the original high-Q recurrence

The final Phase-2 interpretation is not that the numerical outputs were “wrong”. The outputs accurately record the implemented algorithm and its diagonalization convention.

What changed is the evidential interpretation.

The high-Q coordinate recurrence is explained by the conjunction of:

1. exact normalized spectral-rank preservation under $k \mapsto 2k + 1$;
2. exponentially growing spectator multiplicity in the fixed five-term Pauli algebra;
3. candidates lying on repeated-algebra boundaries while the frozen local controls move increasingly deep inside repeated blocks;
4. a residual REAL – NULL pair score that is not invariant under physically equivalent rotations in the full degenerate eigenspaces.

Therefore the original five-term high-Q protocol does *not* establish transport of an intrinsic physical Soft-Space hotspot between independently generated Hilbert spaces.

This conclusion is stronger, not weaker, than merely saying that a proof is missing. The original interpretation has an explicit mathematical mechanism that explains the recurrence and an explicit predeclared counterexample to the required basis invariance.

29 What Phase 2 did establish

A fair scientific summary must preserve both the positive and negative results.

29.1 Empirically established within the implemented protocols

- Independent 9Q perturbation tests produced reproducible positive susceptibility and projector-overlap contrasts for several frozen candidates.
- The harmonized high-Q statistic produced locally prominent recurrent ordinal spectral boundaries.
- Both frozen 12Q candidate coordinates passed the predeclared $6/8 = 0.75$ prominence gate.
- A reduced Qiskit gate-level witness preserved positive REAL – NULL contrast in ideal and selected noisy simulation.

29.2 Proved mathematically

- $0 \leq C_i \leq 1$.
- C_i is invariant under basis rotations inside the already selected rank-two subspace.
- The target/control prominence functional satisfies a deterministic Lipschitz stability theorem.
- A decoupled ancilla branch lift preserves the effective operator, coherence ratio, and complete prominence functional exactly.
- Regularized-resolvent and effective-operator perturbation bounds hold under their stated assumptions.
- The controlled off-diagonal ancilla coupling has exact parity, forcing an even prominence response and eliminating the linear term.
- The Schur complement produces an explicit quadratic branch self-energy.
- The controlled frozen positive instances admit an outward-rounded computer-assisted positive-prominence radius of 8.9×10^{-11} .

29.3 Falsified or not established

- The arithmetic rule $k \mapsto 2k + 1$ is not itself a physical Hilbert-space intertwiner.
- The original independently generated 11Q–12Q Hamiltonians are not exact copied ancilla lifts in the audited same-seed coupling.
- The two 12Q candidate confirmations are not independent replications.
- The fixed local controls are not boundary-matched at high dimension.
- The original ordinal two-vector score is not invariant under rotations in larger degenerate eigenspaces.
- The original five-term high-Q recurrence therefore does not establish an intrinsic basis-independent transported hotspot.

30 Why the negative endpoint is scientifically valuable

The strongest outcome of Phase 2 is methodological.

A weak research workflow could have stopped after the visually attractive 12Q confirmation. Instead, the project proceeded through:

empirical recurrence $\rightarrow$ operator definition $\rightarrow$ exact baseline theorem $\rightarrow$ perturbation bounds
 $\rightarrow$ validated certificate $\rightarrow$ ensemble-coupling audit
 $\rightarrow$ sparse-algebra audit $\rightarrow$ basis-invariance falsification.

This chain separates a genuine mathematical theorem from an invalid generalization of that theorem. It also identifies the precise mechanism by which a sparse high-dimensional Hamiltonian model can generate apparently stable ordinal spectral structure without providing a basis-independent physical subspace observable.

From a research-integrity perspective, this is preferable to increasing the number of seeds in a protocol whose physical observable has already failed a necessary invariance condition.

31 Requirements for a renewed physical model

A new experiment should not merely rerun the same five-term ensemble. It must change the scientific model.

Two principled routes are available.

31.1 Route A: complete degenerate-eigenspace projectors

Instead of selecting an arbitrary two-vector slice, use the full spectral projector

$$P_E = \mathbf{1}_{\{E\}}(H)$$

or a physically defined cluster projector

$$P_{\mathcal{I}} = \mathbf{1}_{\mathcal{I}}(H)$$

for a spectral interval $\mathcal{I}$.

The effective operator becomes

$$\Sigma_P^{(\eta)} = PVQ g_{\eta}(E_P - QHQ) QVP,$$

with P now invariant under any basis rotation inside the degenerate eigenspace.

Controls must be matched by the same algebraic class: boundary projectors should be compared with boundary projectors, not with interior slices in flat repeated blocks.

31.2 Route B: physically richer Hamiltonian ensembles

A second route is to remove the growing spectator degeneracy by allowing the number and locality structure of Hamiltonian terms to scale with q . A generic local model could take the form

$$H_q = \sum_{j=1}^q \mathbf{h}_j \cdot \boldsymbol{\sigma}_j + \sum_{\langle j,k \rangle} \sum_{\alpha, \beta} J_{jk}^{\alpha\beta} \sigma_j^{\alpha} \sigma_k^{\beta} + \dots$$

with a number of independent coefficients that grows with system size.

The design requirements should include:

1. predeclared ensemble family;
2. predeclared candidate observable;
3. basis invariance by construction;
4. boundary-matched control geometry;
5. independent discovery and holdout seeds;
6. effective sample size and eligibility reporting;
7. dependence-aware statistics for multiple candidate chains;
8. frozen thresholds before holdout execution.

32 Statistical considerations

The Phase-2 decision gate was deliberately simple:

$$\hat{p} \geq 0.75.$$

For $B = 8$, this means at least six positive batches. Such a gate is useful as a predeclared operational rule, but it should not be confused with a full hypothesis test.

For independent Bernoulli signs under $p = 1/2$,

$$\Pr[X \geq 6] = \sum_{j=6}^8 \binom{8}{j} 2^{-8} = 0.14453125.$$

In the actual 12Q data, the two candidate chains were highly correlated, so treating them as independent would exaggerate evidence.

A future protocol should therefore use either:

- a hierarchical model that explicitly represents candidate dependence;
- permutation or randomization tests respecting the seed structure;
- a predeclared family-wise error control;
- or a primary single-candidate holdout with secondary descriptive candidates.

Eligibility induced by the neighbouring-gap gate must also be reported. A batch with two eligible seeds is not statistically equivalent to one with five independent eligible seeds.

33 Reproducibility and versioned evidence

The project was developed as a versioned computational audit trail. The important scientific advantage of the long v25.x sequence is not the version number itself, but the preservation of failed and superseded experiments.

The reproducibility package should retain, at minimum:

- every script used for a reported result;
- exact command-line arguments;
- frozen seeds and batch strides;
- output files;
- checkpoint records;
- initial failed coarse interval runs;
- the final Arb/Acb validation output;
- the same-seed coupling audit;
- the sparse-Pauli spectral-artifact audit;
- the degenerate-basis invariance output.

A publication repository should distinguish:

`code/`, `results/`, `documentation/`, `paper/`.

The manuscript should not ask a reader to trust a verbal claim when a frozen machine-readable output exists.

34 Suggested evidence map

Table 7: Phase-2 evidence map.

Stage	Scientific question	Status
v25.2–v25.5	Can 8Q structure guide targeted 9Q search, and can candidate subspaces be tracked under perturbation?	Exploratory-to-confirmatory numerical development.
v25.5.1–v25.7	Do frozen candidates retain more projector overlap or lower susceptibility than spectrum-matched controls on independent seeds?	Several positive candidate-specific numerical results; controls retained.
v25.10–v25.19	Does a small perturbative effective-subspace mechanism recur across targets, families, and local geometry?	Positive numerical evidence; matrix-level mechanism increasingly isolated.
v25.20–v25.30	Does the mechanism transfer across perturbation models and dimensions, and can a frozen rule predict higher-Q coordinates?	Empirical dimensional rule and harmonized frozen validation.
v25.31	Do the two frozen 12Q predictions pass the predeclared local-prominence gate?	Yes: both $6/8 = 0.75$.
v25.32	Can a reduced perturbative witness be expressed and checked at circuit level?	Yes in ideal and selected noisy simulation; not the full 12Q calculation.
v25.33–v25.39	Can dimensional preservation be proved in a controlled non-circular model and certified away from zero coupling?	Yes. Exact lift theorem plus outward-rounded certified positive radius.
v25.40	Is the original independent 11Q–12Q ensemble the controlled ancilla lift?	No in the frozen favourable same-seed audit.
v25.41	Can the coordinate recurrence be explained by the sparse Pauli algebra and spectral-rank geometry?	Yes for the implemented protocol: normalized rank preservation plus growing spectator multiplicity and unmatched controls.
v25.42	Is the residual ordinal two-vector hotspot sign basis invariant inside degenerate eigenspaces?	No. Frozen sign-reversal counterexamples while H is unchanged to $\sim 10^{-14}$.

35 Limitations and non-claims

For clarity, this manuscript does *not* claim any of the following:

1. that the original five-term random-Pauli ordinal recurrence is a universal law of finite-dimensional Hilbert spaces;
2. that the 12Q confirmation constitutes two independent replications;
3. that the controlled ancilla theorem proves the independently redrawn 9Q–12Q ensemble;
4. that the Qiskit reduced witness is equivalent to the complete 12Q classical prominence computation;
5. that the computer-assisted radius is a confidence interval for a random distribution of Hamiltonians;
6. that an ordinal rank-two eigenspace slice inside a larger degeneracy is physically meaningful without an additional basis-selecting observable.

These non-claims are not editorial qualifications added after the fact. They are the logical consequences of the final audit and are part of the scientific result.

36 Discussion

Three conceptual lessons emerge.

36.1 A recurring coordinate need not be a transported state

The identity

$$\frac{2k+2}{2^{q+1}} = \frac{k+1}{2^q}$$

shows how an integer recurrence can simply preserve normalized spectral position. Without an operator intertwiner,

$$H_{q+1}J = JH_q,$$

there is no reason to interpret that recurrence as transport of an eigenstate or eigensubspace.

36.2 A good local score still needs a physical domain

The normalized coherence ratio is mathematically nontrivial and useful. Its effective operator has a clean Feshbach structure, a bounded range, and exact internal $U(2)$ invariance. None of those properties guarantees that an ordinal pair is a physically well-defined object when embedded inside a larger degenerate eigenspace.

This is a general lesson in computational spectral physics: invariance of a formula with respect to coordinates inside a chosen subspace does not imply invariance of the *subspace-selection rule* itself.

36.3 A falsification can preserve the valuable mathematics

The basis-invariance failure does not invalidate:

- the projection-operator derivations;
- the exact decoupled-ancilla theorem;
- the parity theorem;
- the Schur-complement mechanism;
- the interval-arithmetic certificate.

It invalidates a proposed identification between those results and the original independently generated sparse high-Q ensemble.

This separation is exactly why the proof ledger is useful.

37 Conclusion

Phase 2 began by asking whether Soft-Space candidates discovered in low-qubit Hilbert spaces could be continued into 9Q–12Q and whether their apparent stability could be given a physical and mathematical interpretation.

The empirical programme succeeded in constructing a reproducible continuation protocol and culminated in a frozen 12Q result in which both predicted candidate coordinates passed the predeclared 0.75 local-prominence gate. A perturbative matrix mechanism was isolated, expressed as a regularized Feshbach-type effective operator, and translated into a robust REAL – NULL prominence statistic.

The mathematical programme then went significantly further. It proved an exact non-circular dimensional-lift theorem for a controlled ancilla model, derived perturbation and prominence bounds, identified an exact parity selection rule, reduced the controlled coupling by a Schur complement, and produced an outward-rounded computer-assisted certificate for the frozen positive instances.

The final falsification programme established the boundary of those positive results. The independently generated high-Q Hamiltonians do not satisfy the elementary copied-ancilla intertwiner. The fixed five-term Pauli ensemble develops large exact degeneracies as q grows. The arithmetic recurrence $k \mapsto 2k + 1$ preserves normalized spectral rank, and the chosen local controls become trapped inside repeated blocks. Most decisively, the remaining ordinal two-vector hotspot sign can be reversed by basis rotations inside degenerate eigenspaces while the physical Hamiltonian is unchanged.

Accordingly, the original five-term high-Q recurrence is mathematically exhausted as evidence for an intrinsic transported physical hotspot. The proper Phase-2 endpoint is:

The project has established a substantive perturbative effective-subspace theory and a rigorous controlled dimensional-lift result, while simultaneously falsifying the stronger interpretation that the original sparse-Pauli ordinal high-Q recurrence is a basis-independent physical Soft-Space transport law.

That endpoint defines the next research problem cleanly. A new phase should replace arbitrary ordinal slices by basis-invariant spectral projectors or use a physically richer Hamiltonian ensemble that removes the spectator degeneracy, together with boundary-matched controls, independent holdout data, and a fully predeclared inference protocol.

A Proof-status ledger

Statement	Status
$k \mapsto 2k + 1$ is an exact label map	Proved exactly.
The two branch injections are distinct from the arithmetic source-label map	Proved from the frozen protocol.
$0 \leq C_i \leq 1$	Proved exactly.
C_i is invariant under $U(2)$ rotations inside the already selected rank-two subspace	Proved exactly.
Uniform target/control score bounds imply an explicit prominence bound	Proved under stated assumptions.
Exact decoupled-ancilla lift preserves Σ_i , C_i , and full prominence	Proved exactly for the controlled lift.
$\rho < 1/2$ is sufficient to preserve cluster isolation under the stated gap normalization	Proved as a sufficient condition.
Controlled $X_a \otimes G$ coupling makes prominence even in ε	Proved exactly for the controlled path.
Schur complement gives a quadratic branch self-energy	Proved exactly under the invertibility condition.
Frozen controlled positive instances have common $\rho_{\text{cert}} = 8.9 \times 10^{-11}$	Computer-assisted, outward-rounded certificate for the stored matrices.
Original independent 11Q–12Q Hamiltonians are exact copied ancilla lifts	Refuted for all 40 frozen same-seed audit cases.
High-Q ordinal recurrence can be explained by normalized-rank preservation, sparse-Pauli multiplicity, and control geometry	Established for the implemented five-term protocol.
The two frozen 12Q candidate chains are independent	Refuted in the frozen data; batch-prominence correlation ≈ 0.9894 .
Ordinal two-vector hotspot sign is invariant under basis rotations in larger degenerate eigenspaces	Falsified by frozen finite counterexamples.
Original five-term high-Q protocol demonstrates an intrinsic transported basis-independent hotspot	Not established; the required interpretation is rejected by the final audit.

B Key equations in compact form

$$P_i = |u_i\rangle \langle u_i| + |u_{i+1}\rangle \langle u_{i+1}|, \quad (1)$$

$$Q_i = \mathbb{I} - P_i, \quad (2)$$

$$E_i = \frac{\lambda_i + \lambda_{i+1}}{2}, \quad (3)$$

$$g_\eta(x) = \frac{x}{x^2 + \eta^2}, \quad (4)$$

$$\Sigma_i^{(\eta)} = P_i V Q_i g_\eta(E_i - Q_i H Q_i) Q_i V P_i, \quad (5)$$

$$C_i = \frac{\|\Sigma_i^{(\eta)}\|_F}{\sum_{n \notin \{i, i+1\}} |g_\eta(E_i - \lambda_n)| \|w_n^\dagger w_n\|_F}, \quad (6)$$

$$D_{f,k,b} = \text{median}_r \left(C_{f,k,b,r}^{\text{REAL}} - C_{f,k,b,r}^{\text{NULL}} \right), \quad (7)$$

$$R_{k,b} = \min\{D_{Z,k,b}, D_{X,k,b}\}, \quad (8)$$

$$B_{k,b} = \text{median}_{c \in N_q(k)} R_{c,b}, \quad (9)$$

$$\Pi_{k,b} = R_{k,b} - B_{k,b}, \quad (10)$$

$$\hat{p}_k = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{\Pi_{k,b} > 0\}. \quad (11)$$

The arithmetic identity behind the observed source-coordinate recurrence is

$$k_{q+1} = 2k_q + 1 \implies \frac{k_{q+1} + 1}{2^{q+1}} = \frac{k_q + 1}{2^q}.$$

The exact controlled lift is

$$H^\uparrow = |0\rangle \langle 0| \otimes (H - \Delta \mathbb{I}) + |1\rangle \langle 1| \otimes (H + \Delta \mathbb{I}), \quad V^\uparrow = \mathbb{I}_2 \otimes V,$$

with

$$\Sigma_{i,b}^\uparrow = J_b \Sigma_i J_b^\dagger, \quad C_{i,b}^\uparrow = C_i, \quad \Pi_{i,b}^\uparrow = \Pi_{i,b}.$$

For the structured coupling,

$$H(\varepsilon) = \begin{pmatrix} A & \varepsilon G \\ \varepsilon G & D \end{pmatrix},$$

and

$$\Sigma_A(E, \varepsilon) = \varepsilon^2 G(D - E)^{-1} G.$$

C Recommended figures for the publication version

The long-form manuscript should eventually be supplemented with the following figures generated directly from frozen outputs:

1. Phase-2 pipeline: discovery, perturbative confirmation, dimensional transfer, 12Q confirmation, theorem track, falsification track.
2. Hilbert-space scaling $8Q \rightarrow 12Q$.
3. 9Q projector-overlap and susceptibility confirmation by candidate.

4. Dimensional source-coordinate chains under $k \mapsto 2k + 1$.
5. 12Q target versus local-control prominence across eight batches.
6. Qiskit ideal/noisy reduced witness.
7. Controlled-ancilla prominence versus ρ with the $\rho < 1/2$ isolation region marked.
8. Log–log quadratic-response plot demonstrating parity exponent ≈ 2 .
9. Interval-certified curvature/radius table.
10. Sparse-Pauli repeated-spectrum geometry versus qubit number.
11. Local control locations relative to repeated spectral blocks.
12. Degenerate-basis rotation sign ranges for the v25.42 falsification.

D Reproducibility checklist

For each numerical result included in a final archival release, record:

1. script filename and version;
2. full command line;
3. Git commit hash;
4. qubit number;
5. Hamiltonian term count;
6. seed list, base seed, and batch stride;
7. number of batches and seeds per batch;
8. perturbation family and repetition count;
9. regularizer and gap threshold;
10. target coordinates and all controls;
11. output filename;
12. checkpoint filename if applicable;
13. predeclared decision rule;
14. whether the run is discovery, confirmation, theorem validation, diagnostic audit, or falsification.

E Terminology note

The term *Soft Space* is used operationally in this project. It should not be understood as asserting a new fundamental mathematical object merely because a numerical candidate has a high score. In Phase 2 the term denotes a hypothesized subspace or spectral region whose behaviour is tested against matched controls under specified perturbations.

After the final high-Q audit, the phrase *transported Soft-Space hotspot* must not be used for the original five-term ordinal recurrence without an explicit qualifier that the physical basis-independent interpretation was not established and was contradicted by the degenerate-basis falsification.

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